

SARATH CHANDRA GENTELA

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SUMMARY

AI/ML Engineer with 3+ years of full-time professional software development experience building production-grade AI/ML systems, agentic applications, retrieval systems, data pipelines, and AWS cloud infrastructure. Experienced in system design, reliability and scaling, Agentcore and Strands-style multi-agent frameworks, model training/inference lifecycles, optimization, tool calling, observability, and delivering customer-impacting systems at scale.

EDUCATION

University of Illinois Springfield, Springfield, IL

Master's Degree, Data Analytics, GPA: 4.0/4.0 | August 2022 - December 2024

Relevant Coursework: Data Analytics, Database Systems, Machine Learning, Data Mining, Statistics, Distributed Systems, Algorithms

Osmania University, Hyderabad, India

Bachelor of Science, Computer Science | May 2016 - May 2020

TECHNICAL SKILLS

Languages: Python, TypeScript/Node.js, SQL, JavaScript, Java, C++, C, C#

Frameworks: FastAPI, Fastify, React, React Native, Server-Sent Events, REST APIs, Docker, GitHub Actions, Terraform

AI / ML: LLM orchestration, agentic AI, Agentcore Framework, Strands Framework, RAG, HyDE, BM25, embeddings, Pinecone, Cohere Rerank, Gemini, OpenAI embeddings, Transformers, structured outputs, tool calling, model training, model inference, transformer architecture, optimization techniques

AWS / Data: AWS, Bedrock Agents, ECS Fargate, Lambda, S3, SQS, Step Functions, CloudWatch, PostgreSQL, Aurora PostgreSQL, data processing pipelines, ETL workflows, PostHog, Tableau, Power BI

WORK EXPERIENCE

Spice, Austin, TX

Founding AI Engineer / Data Scientist (AI) | February 2025 - Present

- Designed, developed, and deployed production-grade AskSpice retrieval systems over 10,000+ processed episodes and 1M+ searchable chunks using query analysis, HyDE expansion, dense-sparse retrieval, Pinecone hybrid search, PostgreSQL fetches, Cohere reranking, and durable conversation state.
- Engineered Pulse/Lucid's agentic AI control plane with controlled LLM turns, sessions, runs, run events, context packs, typed tools, SSE progress, artifact events, durable memory, and AWS/Bedrock Agent-backed orchestration aligned with Agentcore and Strands-style multi-agent patterns.
- Built end-to-end AI/data pipelines that process raw audio/video into searchable chunks, timestamps, embeddings, multilingual artifacts, and retrieval-ready datasets; scaled bulk processing to 16 concurrent workers and roughly 20 episodes per minute.
- Improved customer-facing search and browse quality through entity carryover, pronoun resolution, time-aware filtering, recency-sensitive ranking, follow-up routing, repeat avoidance, and source-grounded answer generation.
- Owned SDLC for AI/ML and internal agent systems, including requirements, technical design, code reviews, source control, testing, deployment, observability, reliability hardening, PostHog reporting, and production debugging across long-running workflows.
- Collaborated with product and engineering stakeholders to translate ambiguous agent and retrieval requirements into scalable architectures, measurable launch checks, and clean implementation plans.
- Improved production reliability by hardening retry behavior, model fallback defaults, tool-call validation, run-state persistence, progress diagnostics, and recovery paths for asynchronous AI workflows.

Illinois Department of Human Services, Springfield, IL

Research Assistant, Data Analytics | December 2022 - January 2025

- Built Python/SQL data processing workflows for opioid-related datasets, including data cleaning, validation, feature-style transformations, demographic/regional analysis, and Tableau dashboards for stakeholder-facing decision support.
- Prototyped LLM/NLP knowledge-retrieval workflows for internal data access, translating reporting requirements and structured public-sector datasets into repeatable analysis, retrieval, and evaluation workflows.
- Applied machine-learning and statistical analysis techniques to identify regional patterns, validate data quality, and communicate model/data findings to non-technical program stakeholders.

Sunairiya Technologies, Hyderabad, India

Junior Data Analyst | June 2020 - May 2022

- Developed SQL/Python data pipelines, Power BI dashboards, and machine-learning forecasting models for sales-trend analysis, business performance monitoring, and recurring data-driven planning workflows.
- Built reusable data models and reporting logic that connected business datasets to automated analysis outputs, strengthening production-style experience with structured data, metrics, and model-backed insights.

PROJECTS

Content Chief-of-Staff Agent Harness Platform | try-pulse.ai | 2026

- Built the Pulse/Lucid creator-agent control plane with session state, run lifecycle, event streams, context-pack assembly, controlled tool execution, artifact tracking, review surfaces, durable memory, and AWS/ECS/Lambda/SQS/Bedrock Agent infrastructure.
- Implemented transcript-first search-to-clip workflows connecting channel search, episode context, word-level timestamps, source media, S3 artifacts, and reviewable preview outputs.

Search and Chatbot Systems | thespice.ai | 2025 - 2026

- Built scalable conversational search and chatbot systems using hybrid dense/sparse retrieval, HyDE query expansion, Pinecone indexes, PostgreSQL metadata joins, Cohere reranking, time/entity filters, and multi-turn conversation state over large content datasets.
- Operated indexing workflows that transform raw podcast audio into timestamped chunks, quotes, embeddings, searchable metadata, and multilingual artifacts used by search, feeds, chatbots, and creator workflows.

Fully Automated Financial Agent | github.com/akira231097/arthra-council | 2026

- Built a Python financial agent that scans equities/crypto, collects market/macro/news data, runs multi-analyst council review, executes buys through Robinhood, monitors positions, and applies sell-signal/risk controls.

LLM Inference Engineering | Vizura AI Labs Course Completed | 2026

- Completed hands-on work covering tokenization/decode execution, KV cache, prefix caching, chunked prefill, PagedAttention, continuous batching, FlashAttention, quantization, speculative decoding, and vLLM-style serving.